Lab Exercise: A Comparative Study of Quantum Fourier Transform and Discrete Fourier Transform in Speech Signal Analysis-Synthesis and Compression

# Introduction[1]

This exercise introduces the quantum Fourier transform (QFT) [1] and its application to speech analysis synthesis. The aim of this exercise is to introduce students to quantum computing [2] concepts using a user-friendly computing signal processing environment, namely the object-oriented Java-DSP (J-DSP). This computing exercise starts by covering basic concepts in DFT-based analysis-synthesis using peak-picking. The method of peak-picking was previously introduced in sinusoidal analysis-synthesis [3] and compression of speech. The objectives of the exercise are: a) introduce DFT based analysis-synthesis, b) show how peak picking can be used in speech compression, c) show that the optimality of peak picking in terms of signal-to-noise ratio is based on Parseval's theorem, d) introduce the same concepts using the quantum Fourier transform, e) implement a QFT based analysis synthesis, f) compare the DFT and QFT based compression in terms of SNR and discuss briefly the effects of noise in quantum computing. In the next section, we discuss DFT-based analysis-synthesis.

## DFT-based Speech Analysis-Synthesis and Compression

Discrete Fourier Transform (DFT) is a widely used mathematical tool in digital signal processing. It is used to convert a finite sequence of equally spaced samples of a signal into a series of complex numbers. This process gives us the frequency domain representation of the signal, which can be used to analyze and process the signal in various applications.

The equation of DFT for a sequence of N complex numbers is given as

$$DFT(x_j) = \sum_{j=0}^{N-1} x_j . e^{\frac{-2\pi ijk}{N}}$$

For example, the output of the DFT for a sequence of 4 numbers $x_j$, represented as $X_k$ is :

$$X_0 = \sum_{j=0}^{4-1} x_j \,.\, e^{-2\pi ij(0)/4} = x_0 + x_1 + x_2 + x_3$$

$$X_1 = \sum_{j=0}^{4-1} x_j \,.\, e^{-2\pi ij(1)/4} = x_0 - i.x_1 - x_2 + i.x_3$$

$$X_2 = \sum_{j=0}^{4-1} x_j \,.\, e^{-2\pi ij(2)/4} = x_0 - x_1 + x_2 - x_3$$

$$X_3 = \sum_{j=0}^{4-1} x_j \,.\, e^{-2\pi ij(3)/4} = x_0 - x_1 + x_2 - i.(x_3 - x_1)$$

Where $i = \sqrt{-1}$ is the imaginary unit. This formula calculates the four frequency components present in the signal.

The DFT and its fast implementation using the FFT can be used for signal analysis-synthesis using a

[1] This lab exercise was developed by Aradhita Sharma and Prof. Andreas Spanias in January 2023.

constrained set of DFT components. The constrained or reduced set of FFT components is produced by peak-picking of the magnitude of the FFT components and reconstructing the signal from these selected components. A simple DFT-based algorithm for speech compression is shown in Fig 1.

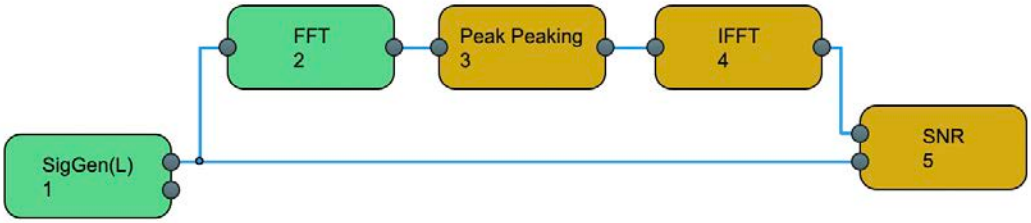


*Figure 1: J-DSP block diagram of DFT-based speech compression using peak picking.*

For example, one can choose say L out of the N components of the FFT based on the magnitude of the frequency components. Based on Parseval's theorem [3], selecting the component with the largest magnitudes will maximize the SNR of this constrained or compressed representation. More sophisticated and perceptually relevant selection of frequency components is based on selecting the harmonics [3]. In addition, high-quality audio representations are based on using more sophisticated perceptual criteria such as those used in MP3 compression [4]. In this exercise, however, we will focus on peak-picking of the FFT for the purpose of speech analysis-synthesis (Fig. 1).

The J-DSP implementation of this method is based on frame-by-frame processing of speech samples. This framing or segmentation function is built in J-DSP in the function **SigGen(L).** To implement the FFT peak-picking we take the following steps to establish the J-DSP simulation in Fig. 1.

### Procedure

1. Establish the J-DSP blocks and form the simulation as shown in Fig. 1.
2. Select one of the speech files in the SigGen(L) block.
3. For each N-point frame, compute the N-point FFT.
4. Using peak picking block, which takes FFT block output as input and compresses the signal,
   - Select the first L components (L<N) and set the rest FFT components to zero.
   - Select the largest L components and set the rest FFT components to zero.
5. Reconstruct the "compressed" speech using the IFFT block with an IFFT size same as the N-point FFT size.
6. Use the SNR block as shown in Fig. 1 and calculate the signal-to-noise ratio in dB, which is the ratio of power of the original signal to the difference of power of original signal and the signal output obtained from the IFFT block.
7. You will observe an increase in SNR as you increase the number of selected L components. If L=N (no compression) the signal reconstruction is as expected perfect.

In the following, we will implement the same process using the QFT and the IQFT. Because of measurement noise, you will observe that the results with the QFT are different than those obtained with the FFT. Before we implement the QFT-based peak-picking, we will give a brief introduction to quantum computing concepts.

# Quantum Computing

Quantum computing is an emerging technology that promises ultra-high speeds that are estimated to be a hundred million times faster than current classical computing. Quantum computing (QC) is based on the principles of quantum theory which describes the behavior of energy and matter on the quantum (atomic and subatomic) level. QC is much faster and is intended to be used to solve a range of complex problems with more efficient algorithms.

QC could help by processing the data faster, and so, has applications in fields like signal processing, data analysis, communications, machine learning, encryption, pattern matching, optimization, etc. [5].

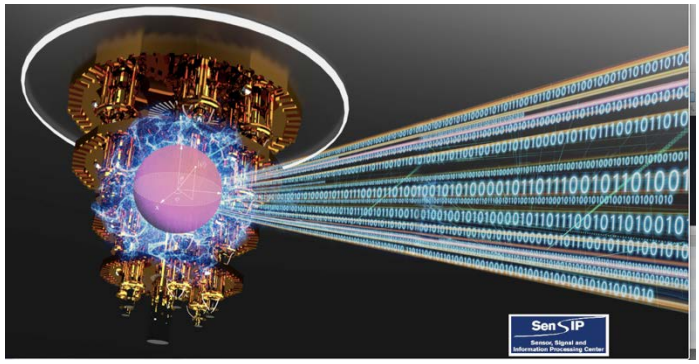


*Figure 2 : Visualization of quantum computer and data processing [6]*

## Quantum Basics

We know, the information contained in a classical bit, which can only be either 0 or 1, is entirely predetermined. However, qubits are written as $|0\rangle$ *or* $|1\rangle$."Qubits" often known as "quantum bits," are the building blocks of quantum computers that can exist in a superposition of both $|0\rangle$ *and* $|1\rangle$ (quantum states expressed as “kets”) at the same time because of the property of superposition of quantum computations. Quantum algorithms are specifically designed to take advantage of the properties of qubits, such as superposition and entanglement, to perform certain tasks faster than classical algorithms. These properties are explained as follows.

Superposition:

Superposition property of quantum mechanics enables the qubit to exist in more than one state. A qubit that is in a superposition of these two states is then written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

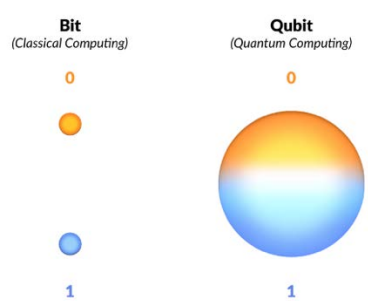


*Figure 3 : Classical bits are represented at the left side, which has two possible states, either be zero or one. Qubit is represented at the right side, which is in the superposition of two states, zero and one, represented as the surface of the Bloch sphere (geometric representation of 2 level quantum system) [7]*

The coefficients $\alpha$ and $\beta$ are called “probability amplitudes”, and they are related to the relative “weighting" of the two states in the superposition. The square of the probability amplitudes sums up to 1, to satisfy the normalization condition

$$|\alpha|^2 + |\beta|^2 = 1$$

In QC, when the value of a qubit is measured, the states are no longer in the superposition and the output is

either only 0 or 1. In quantum mechanics, this is called the measurement problem, where a quantum state exists in a superposition of potential values while not observing it, but it results in only one value when the measurement is done.

Entanglement:

Quantum states can correlate with one another with the help of the entanglement property of quantum mechanics. When two quantum states are said to be entangled, the first state can be determined by the information of second state, that means these states create a single system when they are entangled and interact with one another.

Quantum Noise:

The study of quantum information is fundamentally dependent on quantum uncertainty. Qubits are extremely sensitive to environmental changes, and even a minor alteration can cause the loss of information. All these factors are termed as quantum noise, which alters the measurements in the end [8]. Examples of quantum noise include :

- Depolarizing noise: This type of noise occurs when qubits are affected by their environment and lose their polarization.

- Amplitude damping noise: This type of noise occurs when the qubits lose their amplitude and transition to a lower amplitude state.

- Phase damping noise: This type of noise occurs when the phase coherence of a qubit is affected by its environment.

- Bit-flip noise: This type of noise occurs when the qubits change state due to a physical process such as a magnetic field fluctuation.

- Measurement Error: This occurs during the final measurement of a quantum state. This means that there is always a degree of uncertainty or noise in the outcome of measurement. Measurement error can occur due to bit flip error. For example, if a qubit is in the state $|0\rangle$ and experiences a bit flip error, it may be read out as $|1\rangle$ instead.

Therefore, it is likely that QFT-based simulation is going to have different results compared to DFT-based simulations when quantum noise is present.

## QFT-based Speech Analysis-Synthesis

Quantum Fourier Transform

The quantum Fourier transform (QFT) is the quantum implementation of the discrete Fourier transform over the amplitudes of a quantum wavefunction with a computational complexity of $O(n^2)$ where $n$ is the number of qubits [2]. It is similar to DFT, with the reverse sign of phase exponent. The computational complexity of FFT based DFT implementation is $O(N\log N) = O(n2^n)$ for $N = 2^n$ value. Therefore, quantum computation provides exponential speedup for quantum Fourier transforms. The QFT and the inverse QFT (IQFT) are important algorithms for performing Fourier analysis on quantum states. QFT and IQFT are typically represented mathematically using matrices, which can be implemented as quantum circuits. The QFT and IQFT circuits are defined by equations that specify the operations performed by the quantum gates, which in turn define the quantum circuit. The given equations describe the QFT and its inverse are shown below, and the circuits that are constructed using these equations are shown in the Fig. 4 and 5.

QFT acts on a quantum state $|k\rangle$ and maps it to the quantum output state as :

$$QFT|k\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{2\pi ijk/N} |k\rangle$$

where $i = \sqrt{-1}$. Similarly, IQFT is written as :

$$IQFT|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{-2\pi ijk/N} |j\rangle$$

$$|State\ in\ computational\ basis\rangle \xrightarrow{QFT} |State\ in\ quantum\ Fourier\ basis\rangle$$

$$|State\ in\ quantum\ Fourier\ basis\rangle \xrightarrow{IQFT} |State\ in\ computational\ basis\rangle$$

These mathematical equations that define the QFT and IQFT specify the operations performed by the quantum gates in the circuit, where the Hadamard gate (H, for creating the superposition), two qubit-controlled rotation gate (R, for implementing $e^{2\pi ijk/N}$ rotation), and swap gate (SWAP, for implementing swaps of coefficients mentioned in the example below) are used.

The matrix representation of these gates is :

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},\ R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{\frac{2\pi i}{2^k}} \end{pmatrix},\ R = \begin{pmatrix} I & 0 \\ 0 & R_k \end{pmatrix},\ SWAP = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

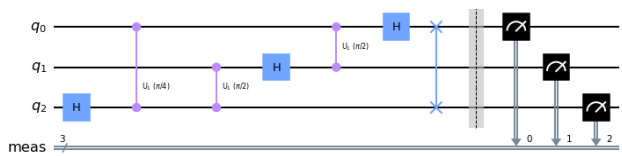


*Figure 4: QFT circuit for 3 qubits after measurement.*

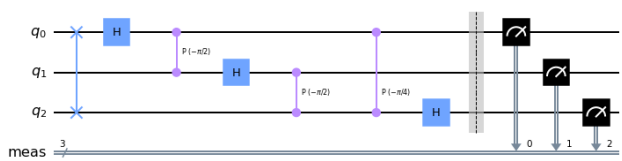


*Figure 5: IQFT circuit for 3 qubits after measurement.*

<u>Normalization</u>

The input signal given to calculate QFT should be normalized in such a way that the square of all the probability amplitudes sums up to 1, to satisfy the normalization condition. Therefore, the input is normalized with norm factor such that :

$$|\alpha_1|^2 + |\alpha_2|^2 + \cdots + |\alpha_j|^2 = 1$$

where $\alpha_j$ are the QFT coefficients and are divided with $\sqrt{N}$ factor and the vector norm, where N = $2^n$, and "n" is the number of qubits [2]. Similarly, normalization is performed for the IQFT function. The normalization factors for QFT and IQFT are shown in the equations below:

$$QFT\ Factor = \sqrt{\mathrm{N}} * vector\ norm$$
$$IQFT\ Factor = vector\ norm/\sqrt{\mathrm{N}}$$

<u>Example of a QFT</u>

For a better understanding of how QFT is calculated, taking an example of a signal $x(n) = \{1,2,0,1\}$, with N = 4, therefore "n = 2" is the number of qubits.

We know, the FFT/DFT of the signal $x(n)$ is FFT$(x(n)) = \{4 + 0j, 1 - 1j, -2 + 0j, 1 + 1j\}$,

For representing the signal as a quantum state, signal is normalized such that $|\alpha_1|^2 + .. |\alpha_n|^2 = 1$

For this normalization, $x(n) = x(n)/vectornorm(x(n))$, where

$$vectornorm(x(n)) = L2norm = sqrt(\ sum(x(n)^2)\ ) = \sqrt{1^2 + 2^2 + 0^2 + 1^2} = 2.45$$

$$modified\ x(n) = \{0.41, 0.82, 0, 0.41\}$$

Representing this normalized signal as a quantum state $|\psi\rangle$ with 2 qubits, we get

$$|\psi\rangle = a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle$$

$$|\psi\rangle = 0.41|00\rangle + 0.82|01\rangle + 0.00|10\rangle + 0.41|11\rangle$$

Calculating the quantum state after the quantum Fourier transform is

$$QFT|\psi\rangle \rightarrow b_{00}|00\rangle + b_{01}|01\rangle + b_{10}|10\rangle + b_{11}|11\rangle$$

where

$$b_{00} = \frac{1}{\sqrt{4}}\sum_{j=0}^{4-1} a_j . e^{2\pi ij(0)/4} = \frac{1}{2}(a_{00} + a_{01} + a_{10} + a_{11}) = 0.82$$

$$b_{01} = \frac{1}{\sqrt{4}}\sum_{j=0}^{4-1} a_j\ .\ e^{2\pi ij(1)/4} = \frac{1}{2}(a_{00} + a_{01}\, e^{i\pi/2} + a_{10}\, e^{i\pi} + a_{11}\, e^{3i\pi/2}) = 0.205 + 0.205j$$

$$b_{10} = \frac{1}{\sqrt{4}}\sum_{j=0}^{4-1} a_j\ .\ e^{2\pi ij(2)/4} = \frac{1}{2}(a_{00} + a_{01}\, e^{i\pi} + a_{10}\, e^{2i\pi} + a_{11}\, e^{3i\pi}) = -0.41$$

$$b_{11} = \frac{1}{\sqrt{4}}\sum_{j=0}^{4-1} a_j\ .\ e^{2\pi ij(3)/4} = \frac{1}{2}(a_{00} + a_{01}\, e^{3i\pi/2} + a_{10}\, e^{3i\pi} + a_{11}\, e^{9i\pi/2}) = 0.205 - 0.205j$$

Normalizing this again, using $QFT\ Factor = \sqrt{\mathrm{N}} * vector\ norm,$ we get

$QFT\ coefficients = b * QFT\ Factor = b * \sqrt{4} * 2.45$ , the QFT coefficients become :

$$b_{00} = 4.01,\ b_{01} = 1.00 + 1.00j,\ b_{10} = -2.009,\ b_{11} = 1.00 - 1.00j,$$

$$QFT|\psi\rangle \rightarrow 4.01|00\rangle + 1.00 + 1.00j|01\rangle + -2.009|10\rangle + 1.00 - 1.00j|11\rangle$$

Here, comparing these coefficients to the one obtained with DFT, it is observed that 2nd and 4th coefficients are swapped, therefore, a swap quantum gate is used at last in the quantum circuit to swap 2nd and 4th coefficients and make the results similar to the DFT. However, the results are not same when these quantum states are affected because of the presence of quantum noise.

<u>Quantum Noise</u>

The performance of quantum computers is substantially impacted by unwanted interactions with the outside environment which show up as quantum noise. Quantum noise is a random process generated because there

are variations in the number of photons that reach the detector from one place to another in quantum computers. The exercise will provide introductory experiences in quantum noise effects including amplitude damping error, and phase damping error. The J-DSP platform also includes measurement error, and LIMA (quantum hardware noise consisting all the noise effects), to explore the effects of noise. These quantum noise gets added to the qubit state according to these equations applied for one qubit:

$$E_{amplitude\ damping} = \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{bmatrix}$$

where $\gamma = sin^2(\theta)$ is the probability of losing a photon.

$$E_{phase\ damping} = \sqrt{1-\alpha}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

where, $\alpha = \frac{1+\sqrt{1-\lambda}}{2} and\ \ \lambda = 1 - cos^2(\varphi)$ is the probability that a photon in the system has been scattered and there is a change in relative phase of the system. This phase damping is the same as the phase flip channel [2].

<u>Peak Picking Process for Efficient Signal Analysis-Synthesis</u>

Same as peak picking is performed for compressing signals using FFT, the data samples are represented in the quantum domain using $2^n$ point, where "n" is the number of qubits using QFT. This can also be extended to 2D signal compression.

For compression, only a small number of QFT components are selected, and the rest components are discarded. After doing this systematically on every frame, the speech signal can be reconstructed using IQFT after compression.

**Procedure**

1. Load the signal from the long signal generator or speech signal frames.
2. For exploring the behavior of real quantum computers, introduce quantum noise.
3. For the signal, take the QFT for different numbers of qubits (will be specified).
4. Normalize the QFT coefficients to be in the frequency domain range.
5. Peak pick some of the frequency components using two different methods.
   - Select the first L components and set the rest QFT components to zero.
   - Select the largest L components and set the rest QFT components to zero.
6. Once the signal's quantum components are modified, reconstruct the signal using IQFT with the same number of qubits as used for QFT, including the effects of quantum noise.
7. Normalize the IQFT coefficients to be in the signal's range.
8. Calculate the Signal to Noise Ratio (SNR) for the original signal and the reconstructed signal using quantum computation.
9. Compress the signal using FFT and IFFT and compare the SNR from classical computation and the SNR obtained using quantum computation.

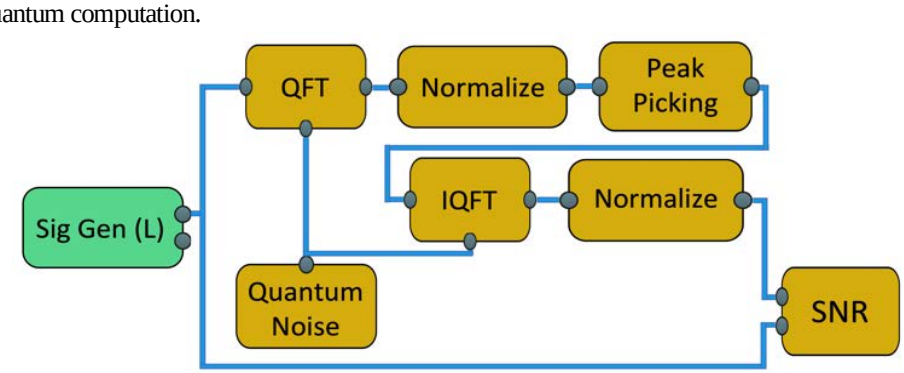

*Figure 6 : Block diagram for QFT-based analysis-synthesis using peak picking including the quantum noise effect.*

# J-DSP Simulation

Perform J-DSP simulation at https://jdspwebsite.netlify.app/. Watch the video on how to set up the block diagram.

Fill out the following table for compression with and without quantum noise effects. Record the SNR value for classical computation and quantum computation and compare the results.

| Quantum Noise & % | Qubits | N point | L components | SNR for Classical Computation | | SNR for Quantum Computation | |
|---|---|---|---|---|---|---|---|
| | | | | SNR (first L) | SNR ( largest L) | SNR (first L) | SNR ( largest L) |
| Amplitude-10 | 6 | 64 | 8 | | | | |
| | 6 | 64 | 16 | | | | |
| | 7 | 128 | 16 | | | | |
| | 7 | 128 | 32 | | | | |
| | 8 | 256 | 32 | | | | |
| | 8 | 256 | 64 | | | | |
| Phase-20 | 6 | 64 | 8 | | | | |
| | 6 | 64 | 16 | | | | |
| | 7 | 128 | 16 | | | | |
| | 7 | 128 | 32 | | | | |
| | 8 | 256 | 32 | | | | |
| | 8 | 256 | 64 | | | | |
| No Noise | 7 | 128 | 32 | | | | |
| | 8 | 256 | 64 | | | | |

**Remarks**

1. Tabulate SNR values.
2. Observe SNR values as L increases with N constant.
3. Observe and compare SNR for classical FFT and quantum FT.
4. Compare the processes of choosing the FIRST vs LARGEST (performance vs compression potential).
5. Observe the SNR values after introducing Quantum Noise (for different types of noise).

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